# Cybersecurity Incident Report

| **Section 1: Identify the type of attack that may have caused this network interruption** |
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| One potential explanation for the website's connection timeout error message is: A **Connection Timeout error** happens when a browser or gateway tries to reach a web server, but the server fails to respond or accept the connection within a set time limit, This usually happens if the web server is completely crashed, offline, or blocked by a firewall<br><br>The logs show that: The logs begin to reflect the web server's struggle to keep up with an abnormal volume of rapid **SYN requests**, The attacker is flooding the system with multiple **SYN requests every second**, overwhelming the server's capacity to respond<br><br>This event could be: **SYN Flood attack** (a type of Denial of Service or DoS attack) |

| **Section 2: Explain how the attack is causing the website to malfunction** |
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| When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:<br>1. The client sends a special packet called a **SYN** (synchronization) to the server<br><br>2. The server gets the SYN packet from the client<br><br>3. The client gets the server's SYN-ACK packet<br><br>Explain what happens when a malicious actor sends a large number of SYN packets all at once: The attacker floods the server with thousands of connection requests, The server responds to each request and sets aside memory to wait for the final confirmation, Because the attacker never finishes the handshake, the server becomes overwhelmed with thousands of these half-open" connections, rapidly draining its system resources<br><br>Explain what the logs indicate and how that affects the server: The network logs show an abnormal, rapid spike in incoming SYN packets arriving every second without any completed connections |